Loop-factor library $\mathcal{L}$

Each row is one parameter-free class of the form $1 \pm c_\sigma$, derived from a Lattice-Defect-FT spinor-trace lemma.

Lemma	Class name	Loop class L_σ	Physical origin
1	**Yukawa-Damping**	$1 \pm \gamma/4$	*$d = 4$ spinor-trace normalization*
2	**PMNS-Self-Energy**	$1 \pm \gamma^2/4$	*double-Wick contraction*
3	**Pure-Self-Energy**	$1 \pm \gamma^2$	*spinor-trace-less vertex*
4	**Resummed-Propagator**	$1/(1 \pm 2\gamma^2)$	*resummation across two chiralities*
5	**Generation**	$1 \pm \gamma/N_{\mathrm{gen}}$	*generation-summed self-energy*
6	**Sub-Generation**	$1 \pm \gamma/(2N_{\mathrm{gen}})$	*$SU(2)_L$ doublet/singlet splitting*
7	**EW-Mixed**	$1 \pm \gamma\,\varepsilon_{\mathrm{sync}}^2$	*Lemma-1 + Goldstone vertex*
8	**Cosmological-Density**	$1 \pm \gamma/2$	*chirality restriction (matter-only)*
--	**Pure-Sync**	$1 \pm \varepsilon_{\mathrm{sync}}^2$	*pure Goldstone-vertex bosonic*